

VaR Backtesting under EWMA and Historical Simulation

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1 Introduction

This report backtests Exponentially Weighted Moving Average (EWMA) and Historical Simulation (HS) Value-at-Risk (VaR) on an equal-weighted portfolio of Apple Inc. (AAPL), JPMorgan Chase & Co. (JPM) and Exxon Mobil Corporation (XOM), spanning technology, financials and energy. Both models generate daily 99% and 95% VaR forecasts, which are formally backtested to assess their statistical accuracy for daily market-risk monitoring and regulatory capital calculation.

2 Data

Daily adjusted closing prices for AAPL, JPM and XOM are sourced from Yahoo Finance via the `quantmod` package in R, from 01 January 2008 to 30 June 2026. This period spans the 2008 Global Financial Crisis and the 2020 COVID-19 shock, making it a demanding sample for VaR backtesting. Prices are converted to daily log returns, and an equal-weighted portfolio (1/3 in each asset) is formed for simplicity and transparency.

All three pairs of assets are positively but imperfectly correlated, with pairwise correlations between 0.34 and 0.48, and the diversification this delivers in the body of the distribution is substantial: the equal-weighted portfolio has a standard deviation of 0.0157 against an average individual standard deviation of 0.0200, a reduction of 21.5%. Whether that benefit survives into the tail, where VaR is measured, is a separate question, and the descriptive statistics in Table 1 bear directly on it.

Figure 1 plots the portfolio return series against the normal distribution. The tails depart from the reference line in both directions and by a wide margin, which is confirmed numerically in Table 1, where kurtosis for all three assets and for the portfolio is well above the value of 3 implied by normality. The portfolio is plotted in preference to the individual assets because it is the series both models are fitted to.

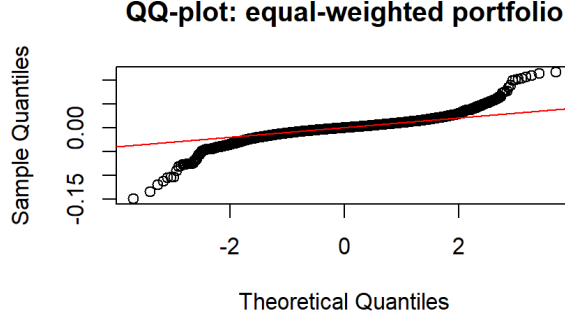


Figure 1: QQ-plot of daily portfolio returns against the normal distribution.

Table 1: Descriptive statistics of daily log returns.

	AAPL	JPM	XOM	Portfolio
mean	0.0008	0.0005	0.0002	0.0005
sd	0.0195	0.0236	0.0171	0.0157
skewness	-0.3395	0.2400	-0.0833	-0.3121
kurtosis	10.3865	19.8656	12.7087	14.6164

Portfolio kurtosis of 14.62 exceeds that of AAPL and XOM individually. Averaging has reduced standard deviation by a fifth without thinning the tail, so the diversification benefit is a property of the body of the distribution and does not carry into the region VaR is designed to measure.

A Ljung-Box test on 20 lags rejects at the 5% level for both returns and squared returns, with statistics of 165.8 and 5623.7 respectively against a critical value of 31.4. With 4,650 observations the test has power to detect autocorrelation far too small to trade on, so the informative comparison is between the two statistics rather than between either and its critical value: the squared-return statistic is 34 times as large. This is the classic signature of volatility clustering, in which the direction of a return is close to unpredictable while the *size* of a return tends to resemble its recent history. It motivates the EWMA and HS models below, both of which adapt their risk estimate to recent volatility.

3 Methodology

Two VaR models are fitted to the portfolio return series. **EWMA VaR** updates conditional variance recursively as $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, using the RiskMetrics decay factor $\lambda = 0.94$, and computes VaR under conditional normality as $\text{VaR}_t = -z_\alpha \sigma_t \vartheta$, where ϑ is the portfolio value; VaR is reported per unit of portfolio value throughout ($\vartheta = 1$), so all figures are in return units. **Historical Simulation (HS) VaR** instead takes the empirical α -quantile of the trailing 300 trading days of realised portfolio returns, with no distributional assumption.

Both models are run at the 99% and 95% levels. The first 300 observations are reserved

for the HS estimation window, so both models are backtested over the same out-of-sample period for a like-for-like comparison.

Two backtests are applied to the daily violation series (a violation occurs when realised loss exceeds the VaR forecast): the **Kupiec (1995)** unconditional coverage test, on whether the observed violation rate matches target, and the **Christoffersen (1998)** independence test, on whether violations cluster in time. Because the two statistics are asymptotically independent, their sum is distributed $\chi^2(2)$ under the joint null of correct coverage *and* independent violations; this joint statistic is reported alongside its components. Clustering is undesirable because it implies the model underestimates risk precisely during periods of market stress, the moments when accurate risk measurement matters most.

4 Results

Table 2 reports coverage results, Table 3 independence and the joint test.

Table 2: Coverage results. VR is observed violations over expected. Testing window 4350 observations.

Model	Exp. rate	Viol.	Obs. rate	VR	99% MC band	LR (cov.)	p
EWMA_99	1.00%	92	2.11%	2.115	[0.64, 1.40]	41.37	<0.001
EWMA_95	5.00%	220	5.06%	1.011	[0.83, 1.17]	0.03	0.862
HS_99	1.00%	59	1.36%	1.356	[0.64, 1.40]	5.02	0.025
HS_95	5.00%	212	4.87%	0.975	[0.83, 1.17]	0.15	0.701

Table 3: Independence and joint tests.

Model	LR (ind.)	p	LR (joint)	p (joint)
EWMA_99	8.40	0.004	49.77	<0.001
EWMA_95	0.77	0.382	0.80	0.672
HS_99	10.55	0.001	15.57	<0.001
HS_95	28.68	<0.001	28.82	<0.001

Figure 2 plots the daily portfolio loss against both 99% VaR estimates, with EWMA violations marked. The EWMA series is the conditional volatility path scaled by $-z_{0.01}$, so its movement also summarises the estimated volatility process.

5 Analysis and Recommendation

Coverage and independence are distinct failure modes with distinct causes, and the coverage test alone identifies neither. HS-95 makes the point. Its violation ratio of 0.975 and Kupiec

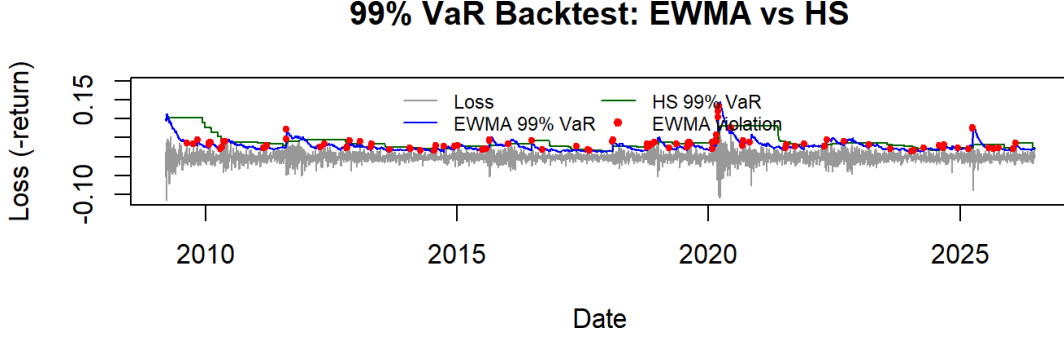


Figure 2: 99% VaR backtest: EWMA versus HS, with EWMA violations marked.

p-value of 0.701 are indistinguishable from a correctly specified model, and the ratio sits well inside the Monte Carlo band of $[0.83, 1.17]$. Yet it records the largest independence statistic in the study, 28.68. The model delivers the right number of violations and places them in the wrong periods. Ranked on coverage it is the strongest of the four specifications; ranked on the joint statistic it is among the weakest.

The two models fail through different mechanisms, and each mechanism determines its remedy.

EWMA fails through its conditional distribution. At the 95% level its violation ratio is 1.011, essentially exact. At the 99% level it is 2.115, against a Monte Carlo band of $[0.64, 1.40]$, 92 violations where the model permits roughly half that number. The Gaussian assumption holds at the 1.645 standard deviations the 95% level requires and breaks down by the 2.326 the 99% level requires, which is what a portfolio kurtosis of 14.62 against the Gaussian benchmark of 3 would predict, and what a skewness of -0.31, which thickens the left tail specifically, reinforces. The coverage failure is confined to the tail and it is a failure of the assumed distribution rather than of estimation: no choice of λ repairs it, because λ governs how quickly the variance responds, not the shape of the tail conditional on that variance.

Coverage is not the only EWMA failure at the 99% level, however. The independence statistic of 8.40 also rejects, at $p = 0.004$. A violation is followed by a further violation on 7.6% of occasions, against 2.0% following a quiet day. The asymmetry follows directly from the recursion. With $\sigma_t^2 = 0.94\sigma_{t-1}^2 + 0.06r_{t-1}^2$, a loss of four standard deviations raises the next day's variance to $0.94 + 0.06(16) = 1.90$ times its previous value, so the VaR forecast rises by roughly 38% in response to a shock that has just demonstrated the previous forecast was far too low. The model under-reacts on exactly the day it most needs to react. Clustering of this kind is a property of the variance dynamics rather than of the innovation distribution, so a distributional correction addresses the coverage failure without necessarily addressing it. EWMA at 99% therefore fails on two counts, and the recommendation below should not be read as a single remedy for both.

HS fails through equal weighting. Each of the 300 observations in the window carries weight $1/300$ irrespective of age, so the quantile estimate can respond to a change in volatility regime only as the window gradually fills with post-shock data. Through a volatility spike

the estimate remains anchored to a calmer recent past, and violations arrive in bursts rather than independently, the mechanism behind independence statistics of 10.55 at the 99% level and 28.68 at the 95% level. Those two figures are not directly comparable in magnitude: 212 violations identify the transition probabilities far more precisely than 59 do, so part of the gap between them reflects the power of the test rather than a stronger effect. The comparison that does carry weight is between EWMA-95 and HS-95, where the violation counts are close (220 against 212) and the independence statistics nonetheless differ by more than an order of magnitude (0.77 against 28.68). The same equal weighting produces the ghost effect in the opposite direction: a large loss drops out of the window abruptly after exactly 300 days, and VaR falls for reasons unconnected to current market conditions, visible in Figure 2 as a step down in the HS line roughly a year after each crisis.

These mechanisms cut across the ranking that coverage alone would suggest. HS does dominate EWMA at the 99% level on coverage, 1.356 against 2.115, but not on independence, 10.55 against 8.40; both reject, so the comparison is between two failing models rather than between a passing and a failing one. The joint statistic makes this explicit: only EWMA-95 survives, with LR = 0.80 and $p = 0.672$, against 49.77, 15.57 and 28.82 for the remaining three specifications. A single model passes a single confidence level, and it is the level that matters least for capital.

The regulatory measure has since moved. Under the Fundamental Review of the Trading Book the internal-models capital charge is Expected Shortfall at 97.5%, not VaR at 99%, though VaR backtesting is retained as the approval gate. For Gaussian returns the two nearly coincide, and the wedge between them opens as the tail thickens, so a portfolio with kurtosis of 14.62 is penalised more heavily under ES than the results above imply.

Recommendation. The two failures require different corrections, and they are complements rather than alternatives: neither model is repaired by the other’s fix.

The EWMA tail failure at 99% is a distributional mis-specification, so the correction is distributional: replace the Gaussian innovation with a Student- t , either as t-EWMA or GARCH- t . This targets the tail directly and leaves the accurate 95% behaviour intact. It does not address the clustering above, which needs a variance specification responding faster to large shocks than a fixed λ of 0.94 permits.

The HS clustering failure is a weighting failure, so the correction is to reweight rather than to lengthen the window. Filtered Historical Simulation standardises returns by EWMA volatility, takes the quantile of the standardised residuals and rescales by current volatility. This preserves the non-parametric tail while restoring adaptivity, addressing both the clustering and the ghost effect with one change.

Until these are implemented, neither model is adequate as the sole basis for capital at the 99% level. In the interim, report the joint statistic alongside the violation ratio in the monthly monitoring pack, and size any capital buffer against the clustering in Table 3 rather than the average violation rate, which Table 2 shows can be correct while the model is failing. A firm running this suite today would see one green light, and it would be on the metric that matters least for capital.